

CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

Dmitry Kurochkin

Spring 2026
Lecture 6

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Matrix Algebra: Definition of Matrix

Def.

A rectangular array A of $n \times m$ numbers

$$A = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1m} \\ A_{21} & A_{22} & \dots & A_{2m} \\ \vdots & & \ddots & \\ A_{n1} & A_{n2} & \dots & A_{nm} \end{bmatrix}$$

is called *matrix*.

Given i, j , the number A_{ij} is referred as (i, j) -th entry of matrix A . Sometimes matrix A is also denoted by $[A_{ij}]_{n \times m}$.

Matrix Algebra: Terminology

Terminology:

An $n \times m$ matrix A is called

- *square matrix* if $n = m$
- *symmetric matrix* if $A_{ij} = A_{j,i}$ for all i, j
- *diagonal matrix* if $A_{ij} = 0$ for all $i \neq j$
- *upper-triangular matrix* if $A_{ij} = 0$ for all $i > j$
- *lower-triangular matrix* if $A_{ij} = 0$ for all $i < j$
- *triangular matrix* if it is either upper- or lower-triangular
- *transpose* of matrix B (write $A = B^T$) if $A_{ij} = B_{ji}$ for all i, j
- *identity matrix* if it is diagonal with $A_{ii} = 1$ for all i

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Arithmetic Operations on Matrices: Addition

Def.

Given $n \times m$ matrices A and B , the sum is an $n \times m$ matrix defined as

$$C = A + B \text{ with } C_{ij} = A_{ij} + B_{ij} \text{ for all } i, j.$$

Arithmetic Operations on Matrices: Multiplication

Def.

Given $n \times m$ matrix A and $m \times s$ matrix B , the product is an $n \times s$ matrix defined as

$$C = AB \quad \text{with } C_{ij} = \sum_{k=1}^m A_{ik} B_{kj} \quad \text{for all } i, j.$$

Arithmetic Operations on Matrices: Inversion

Def.

Given $n \times n$ matrix A , the inversion is an $n \times n$ matrix A^{-1} such that

$$AA^{-1} = \mathbf{I},$$

where $\mathbf{I}$ is the $n \times n$ identity matrix.

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Expectations and Variances of Random Matrices

Let

$$W = \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_n \end{bmatrix} \quad \text{be an } n \times 1 \text{ random matrix,}$$

then we define

$$E[W] \doteq \begin{bmatrix} E[W_1] \\ E[W_2] \\ \vdots \\ E[W_n] \end{bmatrix} \quad \text{and} \quad \text{Var}(W) \doteq E[(W - E[W])(W - E[W])^T]$$

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Multivariate Linear Regression: Terminology and Assumptions

Terminology:

We assume that $X_{i1}, X_{i2}, \dots, X_{ik}$ and Y_i are random and there is relation

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + u_i,$$

where

- index $i \in \{1, 2, \dots, n\}$
- Y_i is called *dependent variable* or *response*
- $X_{i1}, X_{i2}, \dots, X_{ik}$ are the i^{th} observations on each of the k regressors
- function $\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik}$ is referred as *population regression line*
- β_0 is *intercept* of the population regression line
- $\beta_1, \beta_2, \dots, \beta_k$ is *slope coefficient* on $X_{i1}, X_{i2}, \dots, X_{ik}$, respectively
- u_i is the *error term*

Multivariate Linear Regression: Terminology and Assumptions

Assumptions:

We assume that there is the following relation between X_i and Y_i :

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + u_i \quad \text{with } i \in \{1, 2, \dots, n\},$$

where

- ❶ *exogeneity* assumption: $E[u_i | X_{i1}, X_{i2}, \dots, X_{ik}] = 0$
- ❷ $(X_{i1}, X_{i2}, \dots, X_{ik}, Y_i)$ is i.i.d. sample drawn from a joint distribution
- ❸ $E[X_{i1}^4], E[X_{i2}^4], \dots, E[X_{ik}^4]$ and $E[Y_i^4]$ are nonzero and finite
- ❹ there is no perfect *multicollinearity*: none of the regressors is a linear function of the other regressors

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- **Multivariate Linear Regression in Matrix Form**
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Multivariate Linear Regression in Matrix Form

Let's consider the following Multivariate Linear Regression Econometric model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + u_i \quad \text{with } i \in \{1, 2, \dots, n\}.$$

Then one can equivalently write:

$$\underbrace{\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}}_{\mathbf{Y}} = \underbrace{\begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}}_{\mathbf{X}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}}_{\mathbf{u}} \quad \text{or } \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where

- matrix $\mathbf{X}$ is called *regressors*
- vector $\mathbf{Y}$ is called *response*

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

OLS Estimators

Claim:

Given Multivariate Linear Regression Econometric model,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad \text{with no multicollinearity,}$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & & & \ddots & \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}, \quad \text{and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix},$$

the OLS estimator of $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_k]^\top$ that minimizes the SSR,

$$S(\boldsymbol{\beta}) = \sum_{i=1}^n u_i^2 = \mathbf{u}^\top \mathbf{u}, \quad \text{is given by } \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Unbiasedness of the OLS Estimators

Claim:

Given Multivariate Linear Regression Econometric model,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u} \quad \text{with no multicollinearity and } E[\mathbf{u}|\mathbf{X}] = 0 \text{ (exogeneity),}$$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & \dots & X_{2k} \\ \vdots & & & \ddots & \\ 1 & X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix}, \quad \text{and } \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix},$$

the OLS estimator $\hat{\boldsymbol{\beta}}$ is unbiased:

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}.$$

Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$

Claim:

Given Multivariate Linear Regression Econometric model,

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{u} \quad \text{with no multicollinearity and } E[\mathbf{u}|\mathbf{X}] = 0 \text{ (exogeneity),}$$

the conditional variance of the OLS estimator, given $\mathbf{X}$, is

$$\text{Var}(\hat{\beta}|\mathbf{X}) = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\mathbf{u}\mathbf{u}^T|\mathbf{X}] \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \text{ (sandwich form).}$$

Proof:

$$\begin{aligned} \text{Var}(\hat{\beta}|\mathbf{X}) &= E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T|\mathbf{X}] \\ &= E\left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u} ((\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u})^T \mid \mathbf{X}\right] \\ &= E\left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u} \mathbf{u}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mid \mathbf{X}\right] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\mathbf{u}\mathbf{u}^T|\mathbf{X}] \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1}. \end{aligned}$$

Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$

Note:

Under the *homoskedasticity* assumption,

$$\text{Var}(\mathbf{u}|\mathbf{X}) = \text{E}[\mathbf{u}\mathbf{u}^\top|\mathbf{X}] = \sigma^2\mathbf{I},$$

where $\mathbf{I}$ denotes the $n \times n$ identity matrix, we have:

$$\begin{aligned}\text{Var}(\hat{\beta}|\mathbf{X}) &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{E}[\mathbf{u}\mathbf{u}^\top|\mathbf{X}] \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \sigma^2 \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.\end{aligned}$$

Consistency of the OLS estimator $\hat{\beta}$

Claim:

Given Multivariate Linear Regression Econometric model,

$$\mathbf{Y} = \mathbf{X}\beta + \mathbf{u} \quad \text{with no multicollinearity and } E[\mathbf{u}|\mathbf{X}] = 0 \text{ (exogeneity),}$$

the OLS estimator $\hat{\beta}$ is consistent:

$$\hat{\beta} \xrightarrow{P} \beta \text{ as } n \rightarrow \infty.$$

Proof:

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\beta + \mathbf{u}) \\ &= \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{u} \\ &= \beta + \left(\frac{\mathbf{X}^\top \mathbf{X}}{n} \right)^{-1} \frac{\mathbf{X}^\top \mathbf{u}}{n} \xrightarrow{P} \beta + (E[\mathbf{X}^\top \mathbf{X}])^{-1} E[\mathbf{X}^\top \mathbf{u}] = \beta.\end{aligned}$$

1 Matrix Algebra

- Definition of Matrix and Terminology
- Arithmetic Operations on Matrices
 - Addition
 - Multiplication
 - Inversion
- Expectations and Variances of Random Matrices

2 Multivariate Linear Regression Model

- Terminology and Assumptions of the Multiple Regression
- Multivariate Linear Regression in Matrix Form
- Distribution of Least Squares Estimators
 - OLS Estimators
 - Unbiasedness of the OLS Estimators
 - Conditional Variance $\text{Var}(\hat{\beta}|\mathbf{X})$
 - Consistency of the OLS estimator $\hat{\beta}$
- Omitted Variable Bias

Omitted Variable Bias

Assume that the “true” population line is given by

$$Y = X\beta + q\gamma + u,$$

where variables (X, q) are exogenous, i.e. $E[u|X, q] = 0$, but we estimate β via OLS using X data only:

$$\tilde{\beta} = (X^T X)^{-1} X^T Y.$$

Then

$$\begin{aligned}\tilde{\beta} &= (X^T X)^{-1} X^T Y \\ &= (X^T X)^{-1} X^T (X\beta + q\gamma + u) \\ &= \beta + (X^T X)^{-1} X^T q\gamma + (X^T X)^{-1} X^T u\end{aligned}$$

and hence

$$E[\tilde{\beta}|X, q] = \beta + \underbrace{(X^T X)^{-1} X^T q\gamma}_{\text{Bias of } \tilde{\beta}}.$$